Materials of 3D Printed Bagpipes

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Introduction

Our goal was to find a suitable substitute by researching different filament plastics and finding one with material properties best suited for our needs. When 3D printing something to be used like bagpipes, the specifics about the material are important for a variety of reasons. Metal, for instance, would create an arguably worse sound in the drones, and rust easily if used as a blowpipe. Wood, specifically African blackwood, is one common material that people have used to make the pipes of the bagpipe. The other is a plastic called Polypenco, chemically known as polyoxymethylene. Poly pipes are advantageous in the rain and different environments, while wood allegedly has a higher quality sound. Because we are 3D printing, we needed to find a suitable material that would be functional and realistic for a bagpipe.

Potential Filaments

PLA (polylactic acid) •Food safe •Brittle •Common	**ABS (Acrylonitrile Butadiene Styrene)** •Tough •High temperatures •Warps during printing •Harmful fumes when printing
Carbon Fiber– additive •Improved durability in ABS •Increased chance of clogging •Brittle •Less warping in ABS	**PC (polycarbonate)** •Super strong and durable •Naturally transparent •Hygroscopic (will warp from water in the air while printing)
HIPS (high impact polystyrene) •Strong and flexible •Excessive warping •Harmful fumes when printing	**PVA (polyvinyl alcohol)** •Water soluble •Warps
PETG (Polyethylene terephthalate glycol-modified) •Durable •Poor adhesion, strings in space	**PET (Polyethylene terephthalate)** •High temperature •Rarely used without modifiers

Fig 1: A list of potential filaments and their benefits and detriments from comparison websites. PETG, highlighted, is what was used.

The requirements and preferences were for a filament type that could withstand some force, not warp under the heat of printing, not absorb or warp with water, and that the available printers could print. Filaments like ABS and HIPS require a ventilation system on the printer due to toxic fumes, which was not available for this project.

Material Properties

Most material properties of each filament depended significantly on the manufacturer that the filament comes from. Focusing on PETG filament, the average printing temperature is 240°C, consistent through manufacturers. The tensile yield strength varied drastically through manufacturers, but some previous studies created stress-strain diagrams for one of each type of filament, shown on the left.

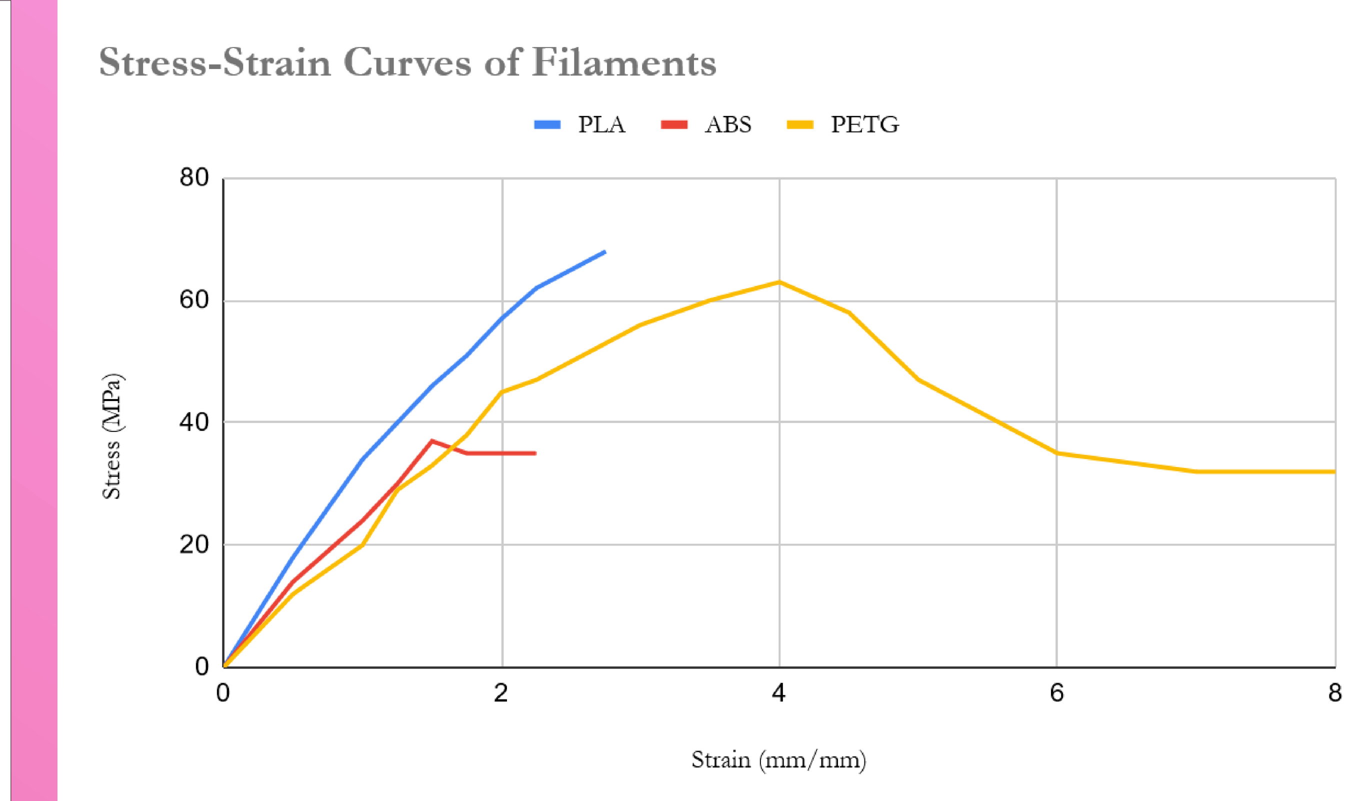
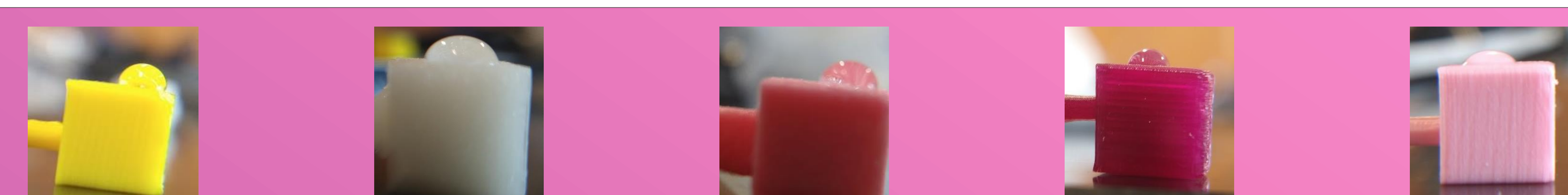


Fig 2: Stress-strain diagram showing the bending of a material, or its strain, in response to additional stress.

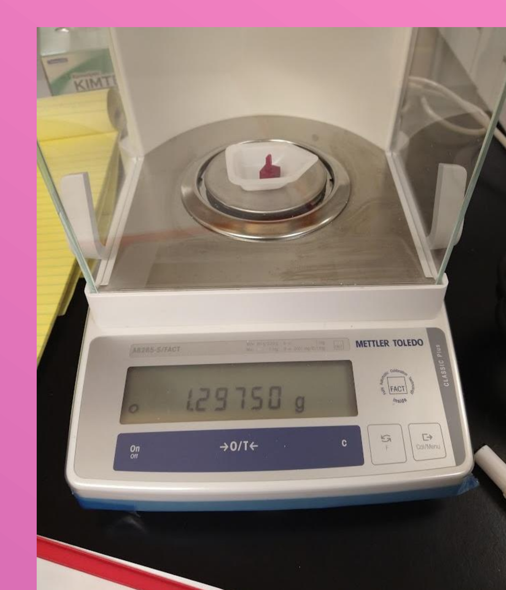
Water repulsion is an important factor for the blowpipe, through which spit and moisture goes with the player's breath. There being no clear consensus of the water repulsion capabilities of the filament, the amount of water taken in was theorized to be related to the settings used to print the pipes, such as the layer height. Each filament used in the pipes was tested against two PLA filaments of similar layer thickness and density.



The test involved printing these boxes at the same settings as the pipes and dropping water on them from an eyedropper. The angle that the drop makes on the surface was expected to correlate to the material's retention of water. From left to right, PLA filament from Hatchbox and Anycubic, and PETG from Hatchbox, Prusa, and Polymaker.

PLA	Anycubic (glow i	Hatchbox (yellow)		Hydrophobicity:
Contact angle (print-aligned)	79	114		A contact angle over 90 degrees (100%) is considered hydrophobic
Contact angle (print-antialigned)	52	80		
Contact angle (bottom)	58	95		
Hydrophobicity	70%	107%		
Weight gain (g)	0.0021	0.0008		
PETG	Hatchbox Pink	Polymaker Pink	Prusa Purple	
Contact angle (print-aligned)	83	59	73	
Contact angle (print-antialigned)	80	48	62	
Contact angle (bottom)	86	55	78	
Hydrophobicity	92%	60%	79%	
Weight gain (g)	0.0017	0.0004	0.0001	

Fig 3: measurements of contact angles, hydrophobicity based on those, and water content for the 5 blocks tested.



PETG material properties (Hatchbox, Polymaker, Prusa)	
Tensile strength 90, 40, 50 MPa	Biodegradability negative
Hydrophobicity 92, 60, 79 %	Recyclability positive
Water retention negligible	Infill density used 70, 70, 85 %
Melting point ~240 °C	

Fig 4: The results of material testing. The specifications that vary based on manufacturer have the measurements of each specific filament in order.

Sanding

PETG plastic tends to string as it prints, leaving excess material inside the pipes. They were sanded using ceramic sandpaper of 60, 80, and 120 grit., then sanded with water and an 800 grit paper to clear the remaining plastic dust.

Conclusions

The pipes made from PETG filament were found to be much weaker than Polypenco or African blackwood pipes. This may be due to the method of preparation itself or the density of the material. However, the filament is still durable enough to create a functional set of pipes. Polycarbonate filament (PC) is a durable filament that was not well researched, and in the future, I would like to explore it and what it can provide in terms of strength and tone quality.

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Acknowledgements

I'd like to thank my professors, Rebecca and Olivia, for the opportunity I had here. I want to thank my teammates as well for their parts in the project and making the time enjoyable. Thanks to my advisor, Dr. Fleenor, for helping me proofread this poster and for teaching me everything I needed for this project.